



Ministry/Organization Name: Ministry of Power

PS Code: 1380

Problem Statement Title: intelligent chatbot to answer queries pertaining to various Maintenance Processes within Substation

Team Name: Coding Zen

Institute Name: Government Polytechnic, Nagpur

Theme Name: Smart Automation

• Problem Statement

Challenges in Substation Maintenance:

- Substation maintenance is complex, demanding rapid response, accurate diagnostics, and streamlined workflows.

Competition Problem Statement:

- Smart India Hackathon: "Intelligent chatbot to answer queries on substation maintenance processes."
- Current approaches may lack efficiency and precision, impacting critical infrastructure reliability.

Why a Chatbot is Essential:

- Maintenance teams face diverse queries, leading to delays and potential oversights.
- An intelligent chatbot offers scalable, quick, and accurate solutions.

Objective & Impact:

- Develop an AI-driven chatbot for prompt and accurate responses to maintenance queries.
- Improve response times, enhance diagnostic processes, and streamline workflows for increased infrastructure reliability.

Solution Overview

Q&N (Question and Answer):

- The chatbot is designed to answer a wide range of questions related to substation maintenance
- Using natural language processing, it understands and responds to textual queries with precision, providing instant solutions.
-

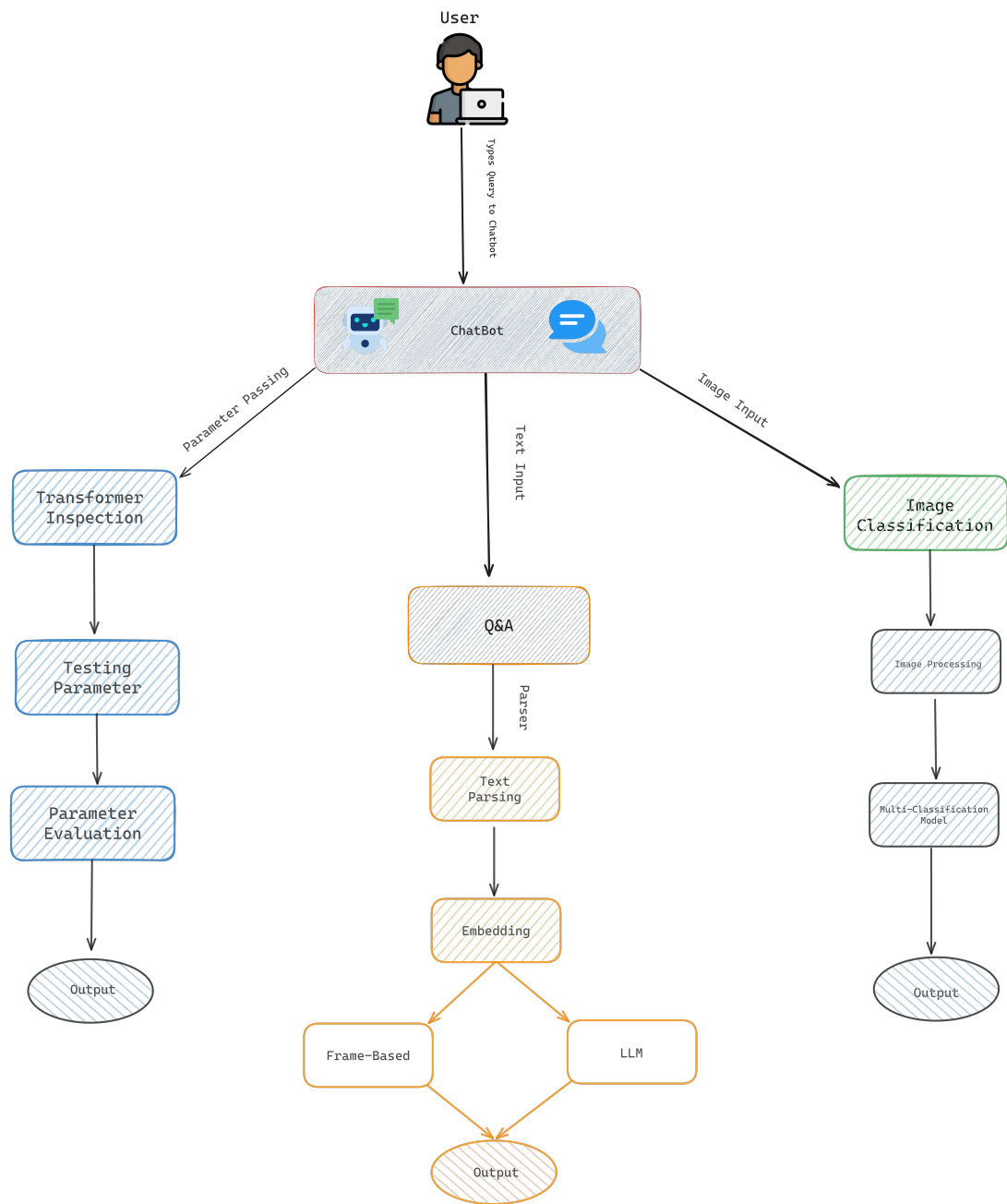
Image Classification

- Advanced image analysis capability for quick problem identification in substation components
- Enables the bot to process images and deliver targeted solutions
-

Transformer Evaluation:

- **Complex Terminology**
Evaluates transformer health by analyzing general parameters
- Proactively identifies potential issues, facilitating preventive maintenance.
-

Flow Chart



Detailed Features - Considering Key Aspects

User Experience:- Text Queries: Intuitive interface for seamless interactions.

- Image Analysis: Easy image upload for quick problem identification, with clear visual feedback
- Testing Functionalities: Comprehensive testing options with detailed feedback.

Mobility:- Mobile-Friendly Design: Responsive on all devices for optimal user experience.

Real-Time Updates: Push notifications for critical issues .

24/7 Accessibility: Anytime, anywhere access for proactive decision-making.

Technology:- Stack: Utilizing Flask, Python, Computer Vision, and AI technologies.

Chatbot:- Frame based Query & LSTM & LLM for Bot.

Image classification:- CNN & Alex Net

Novelty and Complexity:- Novelty: Image analysis for maintenance insights, real-time interactive testing.

Complexity: Multi-layered AI algorithms for diverse scenarios, dynamic Learning for continuous improvement.

Business Model:- Target Audience: Utility companies, maintenance teams in the energy sector.

Cost savings and improved maintenance efficiency.